WhatsApp AI Agent — Full System Design & Business Model

Lebanese SMB Market | Solo Developer Product

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1. Product Summary

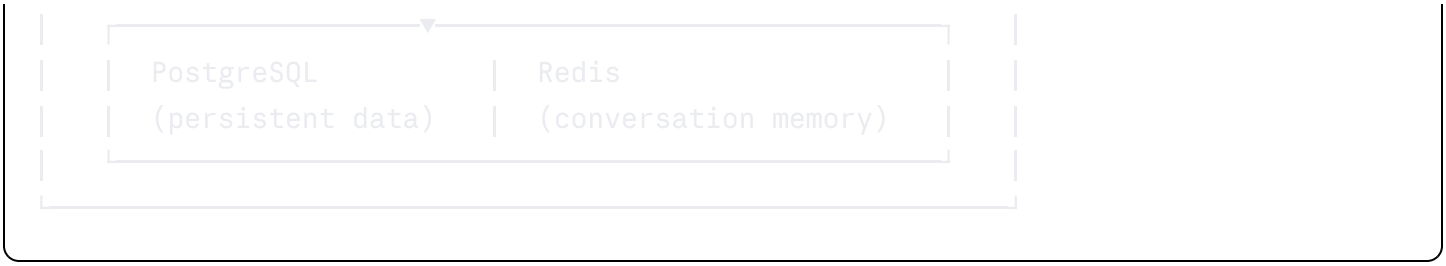
A multi-tenant WhatsApp AI Agent platform sold as a subscription service to Lebanese small and medium businesses. Each business gets an AI agent running on their WhatsApp number that handles customer inquiries, bookings, orders, and escalations — autonomously, 24/7.

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2. System Architecture

2.1 High-Level Overview





2.2 Component Breakdown

n8n — Orchestration Layer

Handles the full agent loop visually. One master workflow serves all clients dynamically.

Responsibilities:

- Receive WhatsApp webhook from Meta
- Extract `business_id` from the incoming phone number mapping
- Call FastAPI MCP to load client config
- Run AI Agent node with the client's system prompt and tools
- Call native n8n tool nodes (Google Calendar, email, WhatsApp reply)
- Call FastAPI MCP for any DB reads/writes
- Send reply back to user via Meta Cloud API

FastAPI MCP Server — Business Logic & Data Layer

A lightweight Python API that acts as the MCP server for the agent. Handles only what n8n cannot natively do.

Responsibilities:

- Load client config (system prompt, enabled tools, escalation contacts)
- Store and retrieve conversation history
- CRUD for orders, appointments, customer records
- Phone number → business_id mapping

PostgreSQL — Persistent Storage

Stores all business and operational data.

Key tables:

- `businesses` — client configs, subscription status, system prompts
- `whatsapp_numbers` — maps phone numbers to business_id

- `conversations` — full message logs per session
- `orders` — order records per business
- `appointments` — booking records per business

Redis — Conversation Memory

Stores short-term conversation context per WhatsApp session (phone number as key). Expires after inactivity (e.g. 2 hours). Prevents the agent from losing context mid-conversation.

Meta Cloud API — WhatsApp Gateway

Free WhatsApp Business API. Receives incoming messages via webhook (POST to your n8n) and sends outgoing messages via HTTP.

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3. Detailed Agent Flow

```
1. User sends WhatsApp message to business number
        ↓
2. Meta sends webhook POST → n8n endpoint
        ↓
3. n8n extracts: user_phone, business_number, message_text
        ↓
4. n8n maps business_number → business_id (via FastAPI MCP)
        ↓
5. n8n loads from Redis: conversation history for this user
        ↓
6. n8n calls FastAPI MCP: get_client_config(business_id)
   Returns: system_prompt, enabled_tools, escalation_number, business_name
        ↓
7. n8n AI Agent node:
   - System prompt = client config
   - Context = conversation history from Redis
   - Available tools = based on enabled_tools flag
        ↓
8. LLM decides:
   a) Reply directly → go to step 11
   b) Call a tool → go to step 9
        ↓
9. Tool execution:
   - Google Calendar → native n8n node
   - Send email → native n8n node
   - Create/get order → FastAPI MCP
```

```
   - Escalate to human → WhatsApp forward via n8n
        ↓
10. Tool result fed back to LLM → loop back to step 8
        ↓
11. Final reply generated
        ↓
12. n8n saves message + reply to Redis (context) and PostgreSQL (log)
        ↓
13. n8n sends reply via Meta Cloud API → User receives message
```

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4. Database Schema

sql

```sql
-- Core business table
CREATE TABLE businesses (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    name VARCHAR(255) NOT NULL,
    industry VARCHAR(100),
    system_prompt TEXT NOT NULL,
    enabled_tools JSONB DEFAULT '[]',
    escalation_number VARCHAR(20),
    subscription_plan VARCHAR(50),
    subscription_status VARCHAR(20) DEFAULT 'active',
    created_at TIMESTAMP DEFAULT NOW()
);

-- WhatsApp number to business mapping
CREATE TABLE whatsapp_numbers (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_id UUID REFERENCES businesses(id),
    phone_number VARCHAR(20) UNIQUE NOT NULL,
    display_name VARCHAR(255),
    is_active BOOLEAN DEFAULT TRUE
);

-- Conversation logs
CREATE TABLE conversations (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_id UUID REFERENCES businesses(id),
    customer_phone VARCHAR(20) NOT NULL,
    role VARCHAR(10) NOT NULL, -- 'user' or 'assistant'
    message TEXT NOT NULL,
    created_at TIMESTAMP DEFAULT NOW()
);

-- Orders
CREATE TABLE orders (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_id UUID REFERENCES businesses(id),
    customer_phone VARCHAR(20),
    customer_name VARCHAR(255),
    items JSONB,
    status VARCHAR(50) DEFAULT 'pending',
    notes TEXT,
    created_at TIMESTAMP DEFAULT NOW()
);
```

```sql
-- Appointments
CREATE TABLE appointments (
    id UUID PRIMARY KEY DEFAULT gen_random_uuid(),
    business_id UUID REFERENCES businesses(id),
    customer_phone VARCHAR(20),
    customer_name VARCHAR(255),
    scheduled_at TIMESTAMP,
    status VARCHAR(50) DEFAULT 'confirmed',
    notes TEXT,
    created_at TIMESTAMP DEFAULT NOW()
);
```

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5. FastAPI MCP Server — Endpoints

```
GET  /client/config/{business_id}          → returns full client config
GET  /client/by-phone/{phone_number}       → returns business_id from phone

POST /conversation/log                     → saves message to DB
GET  /conversation/history/{business_id}/{customer_phone} → returns last N
messages

POST /order/create                         → creates new order
GET  /order/{order_id}                     → fetches order status
PUT  /order/{order_id}/status              → updates order status

POST /appointment/create                   → creates appointment
GET  /appointment/{business_id}/upcoming   → lists upcoming appointments
```

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6. n8n Workflow Design

Master Workflow (Single workflow for all clients)

```
[Webhook Trigger]
      ↓
[Function Node] — parse Meta webhook payload, extract phone numbers
      ↓
[HTTP Request] → FastAPI /client/by-phone/{business_number}
      ↓
```

```
[HTTP Request] → FastAPI /client/config/{business_id}
      ↓
[HTTP Request] → FastAPI /conversation/history (load last 10 messages)
      ↓
[AI Agent Node]
  - Model: Groq / Claude / DeepSeek (swappable)
  - System prompt: from client config
  - Tools:
      • Google Calendar (native)
      • Send Email (native)
      • HTTP Request → FastAPI /order/create
      • HTTP Request → FastAPI /appointment/create
      • HTTP Request → FastAPI /order/{id}
      ↓
[HTTP Request] → FastAPI /conversation/log (save exchange)
      ↓
[HTTP Request] → Meta Cloud API (send reply to user)
```

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7. Infrastructure & Hosting

VPS Setup (Hetzner — recommended)

Service	Spec	Cost
Hetzner CX22 VPS	2 vCPU, 4GB RAM	€4/month
Domain	yourproduct.com	~$10/year
SSL	Let's Encrypt (Certbot)	Free
Total		~$5–6/month

Docker Compose Stack on VPS

yaml

```yaml
version: "3.8"
services:

  n8n:
    image: n8nio/n8n
    ports:
      - "5678:5678"
    environment:
      - N8N_HOST=n8n.yourproduct.com
      - WEBHOOK_URL=https://n8n.yourproduct.com
    volumes:
      - n8n_data:/home/node/.n8n
    restart: unless-stopped

  fastapi:
    build: ./api
    ports:
      - "8000:8000"
    environment:
      - DATABASE_URL=postgresql://user:pass@postgres:5432/agentdb
      - REDIS_URL=redis://redis:6379
    restart: unless-stopped

  postgres:
    image: postgres:15
    environment:
      - POSTGRES_USER=user
      - POSTGRES_PASSWORD=pass
      - POSTGRES_DB=agentdb
    volumes:
      - postgres_data:/var/lib/postgresql/data
    restart: unless-stopped

  redis:
    image: redis:7-alpine
    restart: unless-stopped

  nginx:
    image: nginx:alpine
    ports:
      - "80:80"
      - "443:443"
    volumes:
```

```
      - ./nginx.conf:/etc/nginx/nginx.conf
      - certbot_data:/etc/letsencrypt
    restart: unless-stopped

volumes:
  n8n_data:
  postgres_data:
  certbot_data:
```

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8. Business Model

8.1 Target Market

Lebanese SMBs that:

- Rely heavily on WhatsApp for customer communication
- Have repetitive, high-volume message handling
- Cannot afford a full support team or CRM
- Operate in: clinics, restaurants, retail, real estate, catering, pharmacies

8.2 Pricing Plans

Plan	Price (USD/month)	What's included
Starter	$29	1 WhatsApp number, FAQ handling, escalation, 500 conversations/month
Business	$59	1 WhatsApp number, all tools (booking, orders, follow-ups), 2000 conversations/month
Pro	$99	2 WhatsApp numbers, all tools, unlimited conversations, priority support

Note: Lebanese businesses prefer USD over LBP. OMT/Whish/bank transfer for payment.

8.3 Unit Economics

Item	Cost
VPS (serves all clients)	~$6/month

Item	Cost
LLM API (Groq free tier to start)	$0 during testing
LLM API at scale (DeepSeek V3)	~$0.001 per conversation
Domain + misc	~$2/month
Total infra at 20 clients	~$10–15/month

At 10 clients on Starter plan:

- Revenue: $290/month
- Infra cost: ~$10/month
- **Margin: ~96%**

8.4 Go-To-Market Strategy

Phase 1 — Manual outreach (Month 1–2)

- Pick one vertical (clinics or restaurants — highest WhatsApp pain)
- DM 50 businesses on Instagram/WhatsApp directly
- Offer first month free, no contract
- Goal: 3–5 paying clients

Phase 2 — Referral loop (Month 3–4)

- Ask happy clients for referrals (offer 1 free month per referral)
- Lebanese SMB market is relationship-driven — word of mouth is powerful
- Goal: 10–15 clients

Phase 3 — Content + proof (Month 5+)

- Post case studies on LinkedIn/Instagram showing before/after (response time, missed messages)
- Offer a live demo via WhatsApp — let prospects talk to a demo agent
- Goal: 25+ clients

8.5 Sales Pitch (One Liner)

> "Your WhatsApp is leaking customers. We fix that for $29/month."

8.6 Onboarding Process (Per New Client)

1. Client fills a simple form: business name, products/services, FAQs, escalation number
2. You generate their system prompt from the form
3. Connect their WhatsApp Business number to Meta Cloud API
4. Insert their config row into PostgreSQL
5. Agent is live — no code changes needed

Target onboarding time: under 2 hours per client.

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9. Competitive Advantage

Factor	You	Generic SaaS competitors
Lebanon-specific	✅ Arabic/French/English	❌ English only
WhatsApp-native	✅ Core channel	⚠️ Add-on
Price	$29–99/month	$200–500/month
Local support	✅ Same timezone, same market	❌
Customization	✅ Per-business system prompt	❌ One-size-fits-all

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10. Build Roadmap

Month 1 — Core MVP

- n8n running locally, then on VPS
- FastAPI MCP server with basic endpoints
- PostgreSQL schema deployed
- Meta Cloud API connected
- One working workflow: receive message → LLM reply → send back
- One real client onboarded (free)

Month 2 — Tools + Multi-tenant

- Dynamic client config loading

- Redis conversation memory
- Google Calendar integration (bookings)
- Order creation tool
- Escalation to human
- Second and third clients onboarded

Month 3 — Productize

- Simple admin dashboard (Next.js) for you to manage clients
- Automated onboarding form
- Basic analytics per client (messages handled, escalations)
- Start charging

Month 4+ — Scale

- Client self-service portal (edit FAQs, view logs)
- Stripe/OMT payment integration
- Expand to second vertical